The Pythagorean theorem and its converse

1 X

2 $c = 5$

3
- **a** $(5, 12, 13)$
- **b** 39 units

4 Yes

5
- **a** $c = 50$
- **b** $c = 17$
- **c** $c = 9.49$

6
- **a** $b = 24$
- **b** $b = 21$
- **c** $b = 11.53$
- **d** $b = 14.39$

7 $a = 6$

8 $h = 15$

9 $x = 10.30$

10 $b = 3.00$ mm

11
- **a** $c = 24.04$ cm
- **b** 58.04 cm

12
- **a** $x = 4.90$
- **b** $y = 8.66$
- **c** 13.56

13
- **a** $a = 6$
- **b** $b = 6$
- **c** $x = 8.49$
- **d** 36.49 cm

14
- **a** $x = 5.39$
- **b** $y = 5.39$
- **c** 21 m
- **d** $777

15

a s

b $x^2 + h^2 = s^2$

c Yes

16

a Yes

b No

17

a 15

b

i Obtuse

ii Acute

18

a No, because $19 > 6 + 8$.

b Decrease the side to have a length between 2 and 14.

19

a

i Yes

ii Acute because $12^2 + 9^2 > 13^2$

b

i Yes

ii Obtuse because $8^2 + 6^2 < 11^2$

c

i Yes

ii Right because $8^2 + 15^2 = 17^2$

d

i Yes

ii Acute because $6^2 + 4^2 > 7^2$

e

i Yes

ii Obtuse because $9^2 + 15^2 < \left(3\sqrt{34} + 5\right)^2$

f

i Yes

ii Acute because $7^2 + 5.1^2 > 8^2$

g

i Yes

ii Obtuse because $4.9^2 + 13.2^2 < 16.1^2$

h

i Yes

ii Acute because $4^2 + \left(\sqrt{20}\right)^2 > 5^2$

i

i Yes

ii Obtuse because $\left(\sqrt{131}\right)^2 + \left(\sqrt{337}\right)^2 < 25^2$

j

i Yes

ii Right because $\left(\sqrt{13}\right)^2 + \left(\sqrt{19}\right)^2 = \left(\sqrt{32}\right)^2$

20 $\frac{\sqrt{15}}{2} < x < \sqrt{15}$

21

a $\frac{3\sqrt{2}}{2} < x < \frac{12}{5}$

b $6 < x < 2\sqrt{13}$

Additional questions

22

a $c = 19.24$

b $b = 21.82$

c $b = 9.38$

23 9

24 20.25 ft

25

a $b = 2.24 \text{ ft}$

b $b = 10.25 \text{ in}$

26

a

i $c = 23.46 \text{ in}$

ii 56.56 in

b

i $c = 24.04 \text{ in}$

ii 58.04 in

27 Chenzi incorrectly assumed that the side with length 24 was the hypotenuse when he substituted the values in to the Pythagorean theorem. We can see that $676 - 100 = 576$ as

28 Side length: $s = \frac{15\sqrt{2}}{2}$.

Using the side length to find the perimeter: $P = 30\sqrt{2}$

29

a $x = 20$

b $y = 19.77$

30 $l = 19.72 \text{ m}$

31

a $x = 197.26 \text{ cm}$

b $h = 265.90 \text{ cm}$

32 $h = 5.2$

33 4.93 m

34

a 1083.07 ft

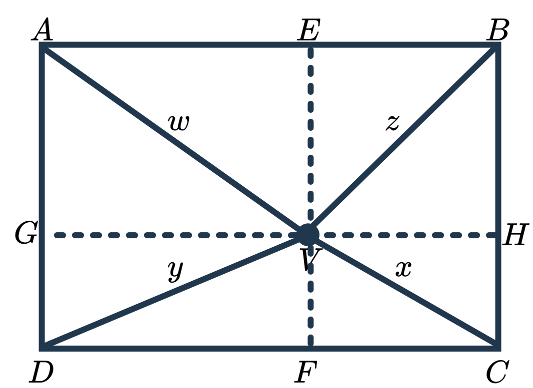
b 1437.92 ft

c Sample Answer: Angelia extends her journey by walking 400 ft south to get dinner. Find the shortest distance between where Angelia started at the beginning of the day and where she ended for dinner. Round your answer to two decimal places.
1799.00 ft

35 Given:

- $ABCD$ is a rectangle
- $\overline{EF} \parallel \overline{AD}$
- $\overline{GH} \parallel \overline{DC}$

Prove: $w^2 + x^2 = y^2 + z^2$



By the Pythagogrean theorem we get the following relationships:

- $w^2 = AE^2 + EV^2$
- $x^2 = VF^2 + FC^2$
- $y^2 = DF^2 + VF^2$
- $z^2 = EV^2 + EB^2$

As $AEFD$ and $EBCF$ are both rectangles w[redacted]ow that $\overline{AE} \cong \overline{DF}$ and $\overline{EB} \cong \overline{FC}$. Using substitution, we get the following relationships:

- $w^2 = AE^2 + EV^2$
- $x^2 = VF^2 + FC^2$
- $y^2 = AE^2 + VF^2$
- $z^2 = EV^2 + FC^2$

Using the above relationships:
$w^2 + x^2 = AE^2 + EV^2 + VF^2 + FC^2$ and $y^2 + z^2 = AE^2 + VF^2 + EV^2 + FC^2$
Using the commutative property of addition:
$y^2 + z^2 = AE^2 + EV^2 + VF^2 + FC^2$
Therefore, $y^2 + z^2 = w^2 + x^2$ by substitution.

36 Notice that $65 = 5 \times 13$ and 5 and 13 are both the largest number of a Pythagorean triple: $\{3, 4, 5\}$ and $\{5, 12, 13\}$.
So this right triangle will either be similar to $\{3, 4, 5\}$ scaled by a factor of 13 or similar to $\{5, 12, 13\}$ scaled by a factor of 5.
Therefore, the other two legs will either be 39 and 52 or 25 and 60.